



NEW MANSOURA UNIVERSITY

Faculty of Engineering · Mechatronics Engineering

Water Flow Meter

Arduino-based real-time flow-rate & volume measurement

using a YF-S201 Hall-effect sensor and an I²C LCD

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Academic Year 2025 / 2026

OUTLINE

What we'll cover

1 Introduction & objective

2 Components & system overview

3 Flow-meter types & why the YF-S201

4 Mechanism, Hall effect & signal

5 Governing equations & worked examples

6 Wiring, pin connections & code

7 Power, calibration & testing

8 Applications, conclusion & references

INTRODUCTION

Turning water flow into numbers

A water flow meter measures how fast water moves through a pipe and how much has passed in total. This project builds one with an Arduino Uno and a YF-S201 sensor, showing the live flow rate and accumulated volume on an LCD.

Objectives

- Measure real-time flow rate (L/h) and total volume (L)
 - Read the sensor with a hardware interrupt — no missed pulses
 - Display results on a 16×2 I²C LCD
- Calibrate against a known volume to keep readings accurate

THE END RESULT



Live readout: flow rate & total volume

How it works — from water to display

Signal chain: from water to display



Key idea: the sensor converts flow into a **pulse frequency**; the Arduino counts those pulses each second to compute flow rate and volume.

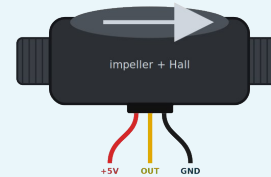
HARDWARE

Components / Bill of materials



Arduino Uno

ATmega328P microcontroller
— the brain that counts pulses
& drives the LCD



YF-S201 Flow Sensor

Hall-effect turbine sensor,
1–30 L/min, pulse output



I²C LCD 16×2

Character display via PCF8574
backpack (address 0x27)



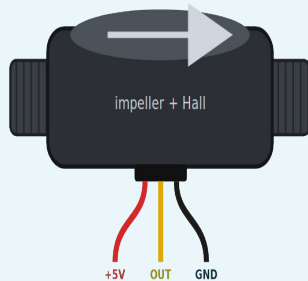
Laptop + USB cable

Powers the board at a steady 5
V through USB

+ Jumper wires · ½" pipe fittings · USB cable for programming

THE SENSOR

YF-S201 — key specifications



3 wires: +5V · signal · GND

Turbine impeller + Hall element, isolated from the water

5–24 V

Operating voltage

≤ 15 mA

Current @ 5 V

1–30

Flow range (L/min)

≈ 450

Pulses per litre

50 ± 10%

Duty cycle

± 3 %

Accuracy (pulse)

f = 7.5 Q

Pulse characteristic

≤ 1.75 MPa

Max pressure

CONTEXT

Types of flow meters



Differential pressure

Orifice / venturi — infers flow from a pressure drop



Positive displacement

Traps & counts discrete fluid volumes (gears / pistons)



Turbine / velocity

A rotor spins with the flow — pulses $\propto$ flow rate

◀ **our YF-S201**



Electromagnetic

For conductive liquids — no moving parts



Ultrasonic

Transit-time / Doppler — clamp-on, non-intrusive

Add a microcontroller → a smart / intelligent flowmeter: digital output, data logging & control (this project).

SELECTION

Why the YF-S201?

Criterion	Why the YF-S201 fits
Fluid	Clean water — well suited to a turbine sensor
Flow range	1–30 L/min covers household taps & demos
Pipe size	½" inline — matches the sensor ports
Accuracy	±3 % (tighter after calibration) — enough here
Installation	Compact & inline; just mind the flow arrow
Cost	Low-cost & widely available — student-friendly

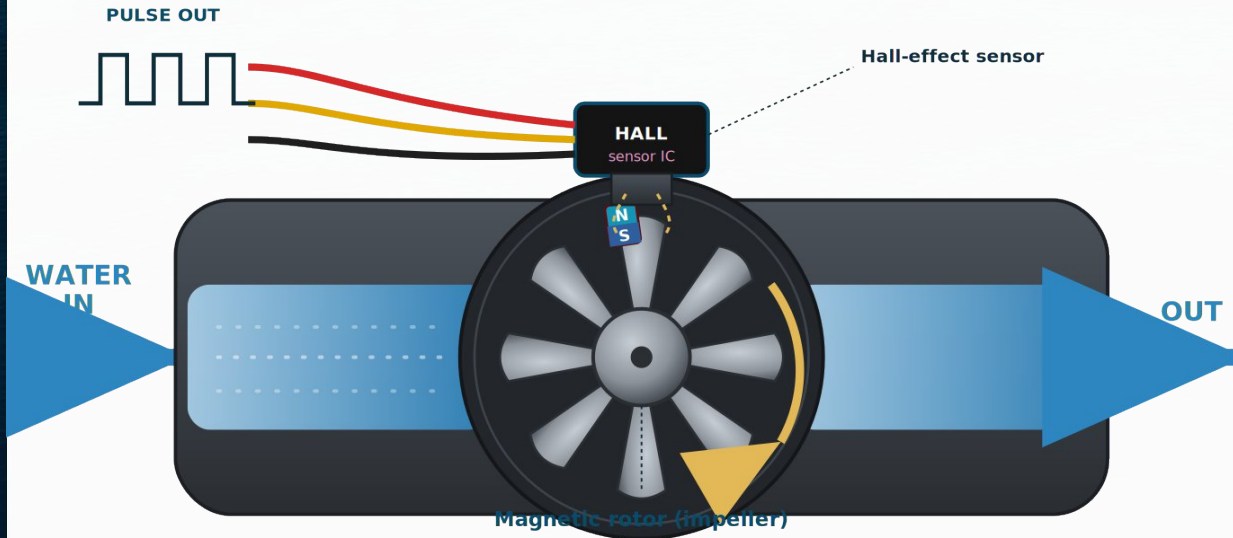
SELECTION FACTORS

- ✓ Fluid properties
- ✓ Flow conditions
- ✓ Accuracy needs
- ✓ Installation
- ✓ Cost / economics

MECHANISM

Inside the sensor — how a pulse is made

Step by step



Flowing water spins the magnetic rotor. Each time the magnet sweeps past the Hall sensor it outputs one pulse → **pulse frequency is proportional to the flow rate.**

- 1 Flowing water spins the turbine rotor
- 2 A magnet on the rotor sweeps past the Hall sensor
- 3 Each sweep produces one clean electrical pulse

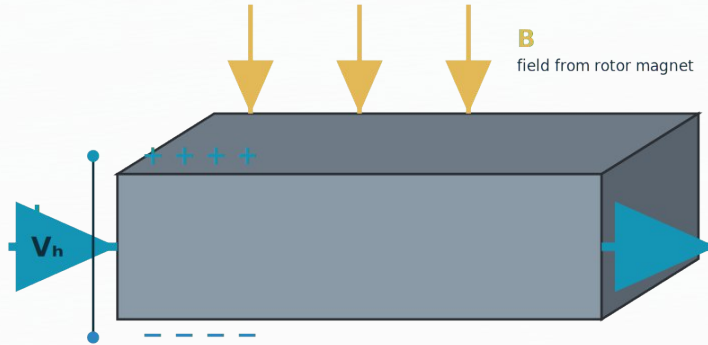
pulses $\propto$ flow rate

THE PHYSICS

The Hall effect behind the sensor

The Hall Effect — how the pulse is born

Why it matters



$$V_h = (I \cdot B) / (n \cdot q \cdot t)$$

n = carrier density q = electron charge t = slab thickness
Magnet sweeps past $\rightarrow B$ pulses $\rightarrow V_h$ pulses $\rightarrow$ one clean pulse out

- A steady current flows through the Hall chip
- The rotor's magnet applies a magnetic field B
- Charges deflect sideways $\rightarrow$ a small voltage V_h appears
- As the magnet sweeps by, V_h pulses $\rightarrow$ a digital pulse out

THE SIGNAL

The output — a train of pulses

Digital pulse output — frequency tracks the flow

Low flow → fewer pulses / second (low f)



High flow → many pulses / second (high f)



$$f \text{ (Hz)} = 7.5 \times Q \text{ (L/min)}$$

The Arduino counts the pulses in one second — **that count is the frequency f.**

THE LAWS

Governing equations

1. Sensor pulse law

$$f = 7.5 \times Q$$

Datasheet: pulse frequency $\propto$ flow (Q in L/min)

2. Flow rate

$$Q_{L/h} = f \times 60 / 7.5 = 8f$$

Convert frequency to litres per hour

3. Accumulated volume

$$V = \Sigma (Q_{L/h} / 3600) \cdot \Delta t$$

Integrate flow over time ($\Delta t = 1$ s each loop)

4. Calibration constant

$$450 \text{ pulses} = 1 \text{ litre}$$

≈ 450 pulses per litre (7.5×60)

Physics behind the pulse: $V_h = I \cdot B / (n \cdot q \cdot t)$

PUTTING THE LAWS TO WORK

Worked examples

EXAMPLE 1

Given: Sensor reads $f = 90 \text{ Hz}$

$$\begin{aligned} Q &= 90 / 7.5 = 12 \text{ L/min} \\ \rightarrow \times 60 &= 720 \text{ L/h} \\ \text{Vol in 1 s} &= 720 / 3600 \end{aligned}$$

$$= 0.20 \text{ L}$$

EXAMPLE 2

Given: Tap runs 6 L/min for 10 s

$$\begin{aligned} f &= 7.5 \times 6 = 45 \text{ Hz} \\ \text{pulses} &= 45 \times 10 = 450 \\ \text{Volume} &= 450 / 450 \end{aligned}$$

$$= 1.00 \text{ L}$$

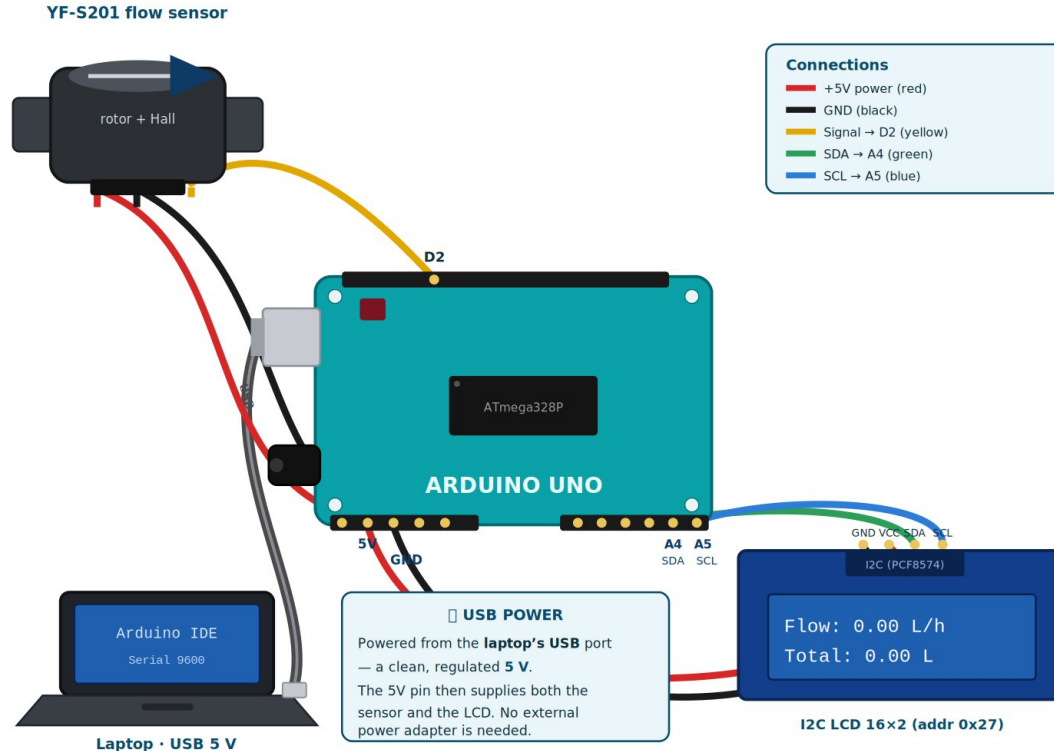
EXAMPLE 3

Given: Dispense 2 L — how many pulses?

$$\begin{aligned} 1 \text{ L} &= 450 \text{ pulses} \\ \text{so } 2 \text{ L} &= 2 \times 450 \\ \text{pulses needed} &= \end{aligned}$$

$$900 \text{ pulses}$$

THE CIRCUIT



At a glance

Sensor

5V · GND · signal→D2

I²C LCD

5V · GND · SDA→A4 · SCL→A5

Power

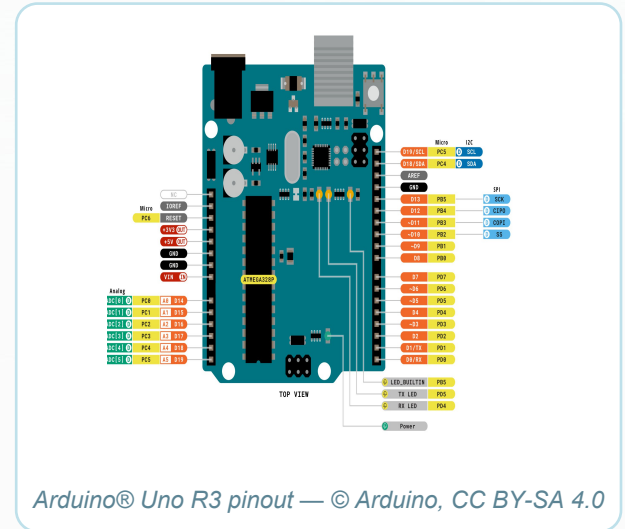
Laptop USB → 5 V

Breadboard-style schematic of every connection

REFERENCE

Pin connections

Wire / signal	Arduino pin	Function
Sensor +5V (red)	5V	+5 V power
Sensor signal (yellow)	D2 (INT0)	Pulse input — interrupt
Sensor GND (black)	GND	Ground
LCD VCC	5V	+5 V power
LCD GND	GND	Ground
LCD SDA	A4	I ² C data
LCD SCL	A5	I ² C clock



Note: D2 is the INT0 interrupt pin; A4/A5 are the I²C lines on the Uno. The board is USB-powered (5 V from the laptop).

The Arduino sketch

```
// D2 = interrupt pin (INT0)
attachInterrupt(
  digitalPinToInterrupt(2), flow, RISING);

void loop() {
  if (millis() >= cloopTime + 1000) {
    cloopTime = millis();
    flow_rate = flow_frequency * 60.0 / 7.5;
    flow_frequency = 0;
    total_volume += flow_rate / 3600.0;
    lcd.print(flow_rate); // Flow L/h
    lcd.print(total_volume); // Total L
  }
}

void flow() { flow_frequency++; }
// ISR: count 1 pulse
```

What the code does

- Every rising edge on D2 triggers flow() — one pulse counted
- The internal pull-up is enabled for a clean, noise-free signal
- Once per second: convert the count to flow ($\times 60 / 7.5$), reset it, add to the running volume
- A hardware interrupt means no pulse is missed, even while the LCD refreshes
- The 7.5 calibration factor comes straight from the datasheet

SETUP & ACCURACY

Power & calibration

POWER · USB 5 V



- Connected to the laptop USB port — a clean, regulated 5 V
- That same 5 V also powers the sensor and the I²C LCD

TOLERANCE & CALIBRATION

The pulse output carries a **±3 % tolerance** — 450 pulses/L (the 7.5 factor) is only the nominal value, so calibrate it to your exact sensor and plumbing.

$$K = \text{pulses counted} / \text{volume (L)}$$

Nominal $K \approx 450$ pulses per litre. Run a known 1 L volume, read the pulses, compute K and set it in the code — a small correction tightens the total-volume accuracy.

ACCURACY

Calibration methods

Calibration = compare the meter to a known standard, then correct it. The standard methods:



Gravimetric

Weigh the fluid collected over a set time — highest accuracy



Volumetric ◀ used here

Collect a known volume in a calibrated tank



Master meter

Compare in series with a trusted reference meter



Prover

A piston / sphere sweeps a calibrated pipe volume

this project: the volumetric method — pour a known 1 L, compare it with the Total, then correct K in the code.

VALIDATION

Testing & results

TYPICAL LIVE READING

Flow: 720.00 L/h

Total: 1.25 L

Updated once every second

≈ 450 pulses = 1 L

Test procedure

- 1 Fit the sensor in-line with the pipe (mind the flow arrow)
- 2 Power up — the LCD shows "Initializing..."
- 3 Run water; the flow rate refreshes every second
- 4 Pour a known 1 L volume and compare with the Total
- 5 Fine-tune the 7.5 factor if the total is slightly off

WHERE IT'S USED

Applications



Water billing

Metering household or building consumption



Irrigation

Precise water delivery in agriculture



Leak monitoring

Spot abnormal or continuous flow



Industrial dosing

Measuring liquids in processes



Beverage machines

Coffee & vending water control



Smart home

Track & automate water use

Conclusion & future work

WHAT WE ACHIEVED

- A working Arduino flow meter using the YF-S201
- Real-time flow rate (L/h) and total volume (L) on an I²C LCD
- Interrupt-driven code — accurate, no missed pulses
- Powered and demonstrated via laptop USB (5 V)

FUTURE IMPROVEMENTS

- Calibrate against a reference meter for higher accuracy
- Add an ESP32 + Wi-Fi for IoT logging & dashboards
- Save totals in EEPROM so they survive a power-off
- Flow alarms + relay for automatic shut-off
- Add a rechargeable 18650 battery pack for untethered field use

References

- 1 YF-S201 Water Flow Sensor — product datasheet, Foshan Shunde Zhongjiang Energy-Saving Electronics Co., Ltd.
- 2 Arduino® Uno R3 full pinout (A000066) — docs.arduino.cc, licensed under CC BY-SA 4.0.
- 3 Arduino libraries used: LiquidCrystal_I2C and Wire (I²C).
- 4 Reference sketch adapted from a public Arduino tutorial and re-documented for this project.
- 5 Hall effect — classical semiconductor physics (E. H. Hall, 1879).

“

Idiots admire complexity;
geniuses admire simplicity.



Thank you

Questions & discussion are welcome

Marrwan · supervised by **Prof. Mohamed M. Tawfik** & **Eng. Yosef Gabr**

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